

7. Indicator species may be used in an investigation to monitor water pollution.

- (a) Describe how you would investigate water pollution at different points along a stream using indicator species. [6 QER]

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(b) The following chart is available to you.

Pollution level**Indicator species****A Clean water**

Stonefly nymph (about 10 mm)



Mayfly nymph (about 20 mm)

B Some pollution

Freshwater shrimp (about 20 mm)



Caddis fly larva (about 10 mm)

C Moderate pollution

Water louse (about 10 mm)



Bloodworm (about 20 mm)

D High pollution

Sludgeworm (about 120 mm)



Rat-tailed maggot (up to 55 mm)

E Very High pollution – no life

Examiner
only

Results from water sampling at two locations are given in the table below.

Species	Total in water sample	
	location 1	location 2
Stonefly nymph	0	0
Caddis fly larva	54	0
Water louse	2	0
Bloodworm	3	0
Sludgeworm	0	80
Rat-tailed maggot	1	65

Use the data in the table to compare the quality of water at the two locations. [2]

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